Fall 2025 - Assignment 1 System Specification & Design Document

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1. Summary:

1.1. System background and goal

This system is a text-based, AI-generated interactive story based on existing tabletop role-playing game (TTRPG) game rule for singleplayer. This system is targeted to experienced TTRPG player and aim to provide similar experience without the need of multiple participants.

A typical TTRPG game consist of one Game master and three to six players. Game master act as the judge and narrator to operate the whole game. His duties are i) Narrative and Storytelling, ii) Rule Adjudication, iii) non-player character (NPC) Management, iv) World-Building and campaign Design. This system uses multiple agents to replaces the game master role in a traditional TTRPG game.

There are some existing products in the market, that utilize the LLM to achieve dynamic storytelling. However, there are some unwanted behaviours which some improvements are targeted to made on this system.

To minimize the workload, this project will focus on the simplified rule of DND 5e, a pre-defined main character, and a short campaign with 3 stages.

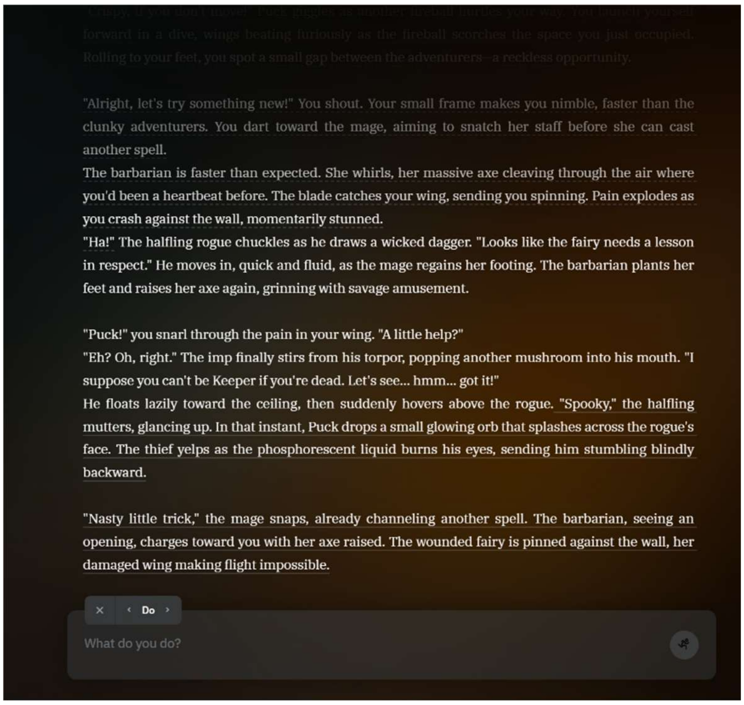


Figure 1. An ongoing game in AI Dungeon.

1.2. Requirements

The baseline features consist of:

1. Dynamic generated story based on user prompt and progress.
2. Interactive control on the main character action.
3. Ensure all action and story align with game rule.

The targeted improvements are:

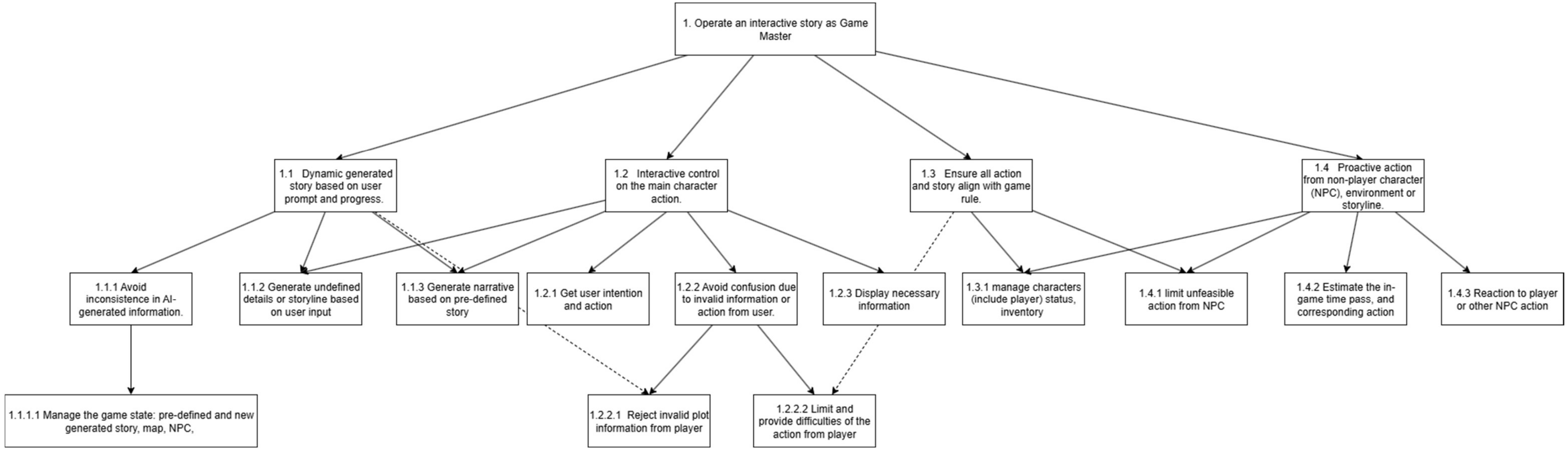
4. Avoid confusion due to invalid information from user.
5. Avoid inconsistence in AI-generated information.
6. Proactive action from non-player character (NPC), environment or storyline.

1. Design Document (MaSE)

The following is the Design Document using MaSE methodology.

1.1. Capturing Goals

1.1.1. Goal Hierarchy



Goal Mapping:

- Interface Role: 1.2, 1.2.1, 1.2.2, 1.2.2.1, 1.2.2.2
- Narrator Role: 1.1, 1.1.1, 1.1.1.1, 1.1.2, 1.1.3
- Judge Role: 1.3, 1.3.1
- NPC Role: 1.4, 1.4.1, 1.4.2, 1.4.3

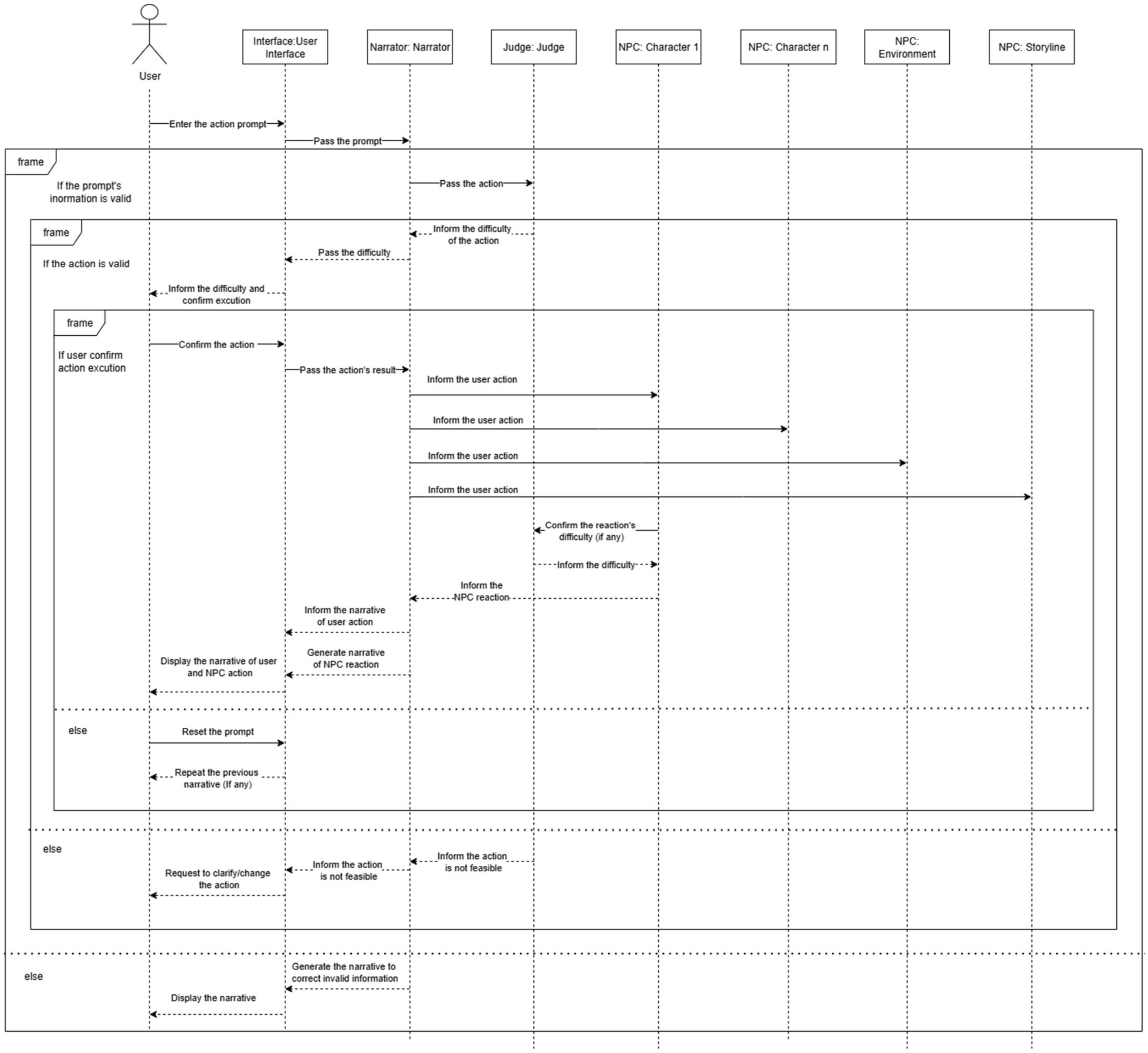
1.2. Applying Use Cases

1.2.1. Use Cases

Brief Description:	User input action or information to the system to progress the story
Precondition(s):	The game started, and the initial message was displayed
Post Condition(s):	The system displays and updates the narrative and the story progress.
Process Steps	
1	The Interface receives the prompt from the user, and passes it to Narrator
2	The Narrator looks in the prompt and identifies the action.
3	Narrator gets the action difficulty from the Judge and pass to interface.
4	The interface displays the difficulty and is pending for player's decision. If continue the action, interface do a random roll to determine the success or not.
5	Narrator generate correspond narrative base on result and inform the NPCs.
6	NPCs evaluate correspond reaction and check its difficulty from the judge
7	NPCs do a random roll to determine the success or not, and info the narrator the result.
8	Narrator generates the NPC's reaction narrative.
9	Narrator passes NPC's reaction narrative and user action narrative to interface
10	The interface shows the narratives to user and progresses the story.

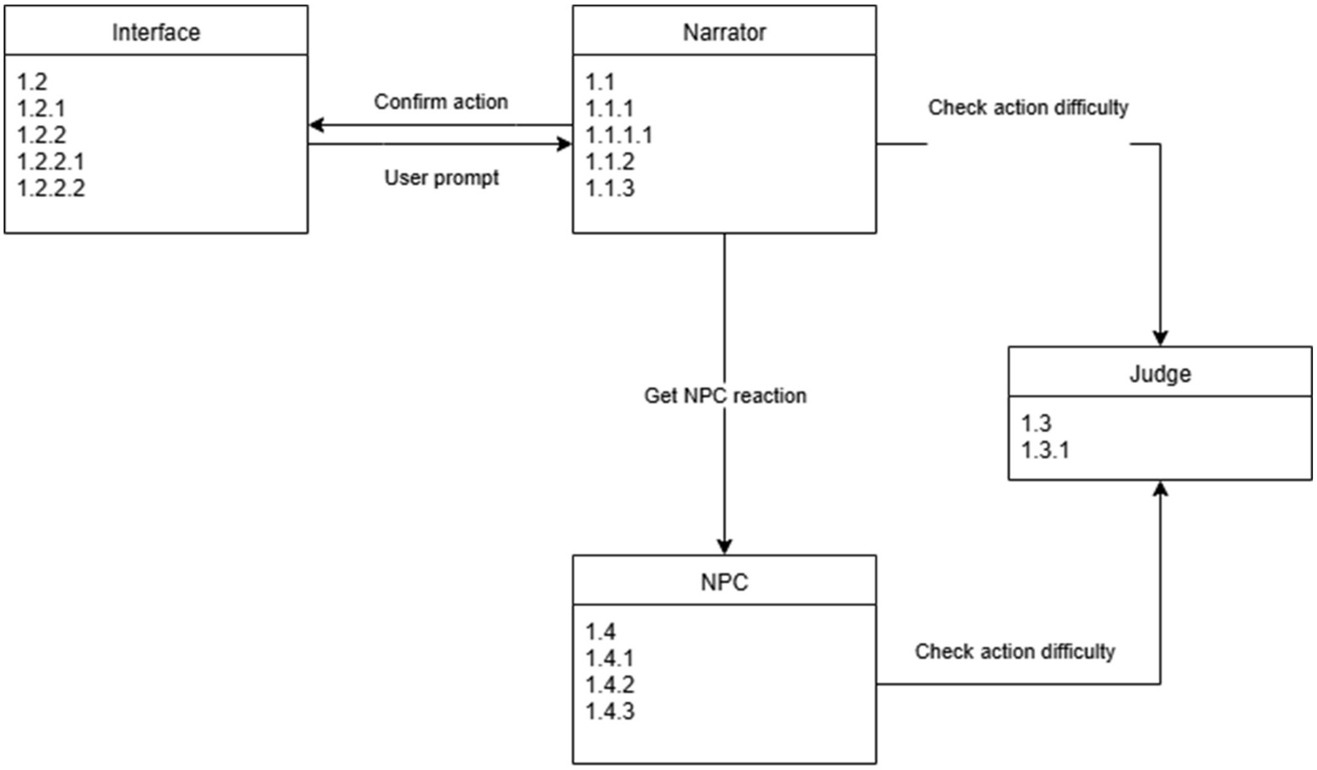
1.2.2. Sequence Diagrams

Use case are applied in the following sequence diagram. There is only one user in this system, but arbitrary NPC instance depends to the story.

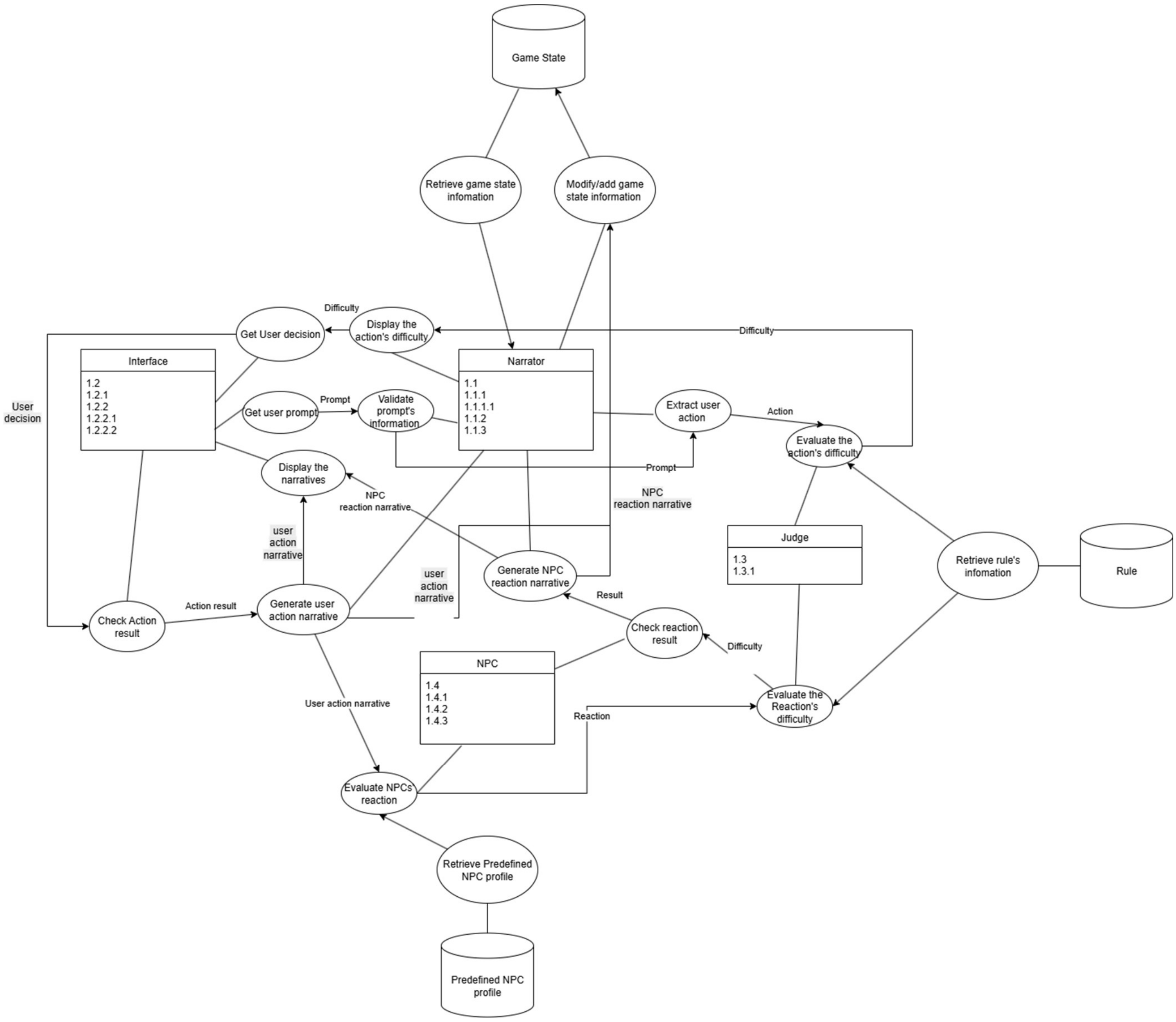

User
Interface:User Interface
Narrator: Narrator
Judge: Judge
NPC: Character 1
NPC: Character n
NPC: Environment
NPC: Storyline
Enter the action prompt
Pass the prompt
frame
If the prompt's inormation is valid
Pass the action
frame
Inform the difficulty of the action
Pass the difficulty
If the action is valid
Inform the difficulty and confirm excution
frame
Confirm the action
If user confirm action excution
Pass the action's result
Inform the user action
Inform the user action
Inform the user action
Inform the user action
Confirm the reaction's difficulty (if any)
Inform the difficulty
Inform the NPC reaction
Inform the narrative of user action
Generate narrative of NPC reaction
Display the narrative of user and NPC action
else
Reset the prompt
Repeat the previous narrative (If any)
else
Inform the action is not feasible
Inform the action is not feasible
Request to clarify/change the action
else
Generate the narrative to correct invalid information
Display the narrative

1.3. Refining Roles

1.3.1. Role Diagram



1.3.2. Role Diagram & Tasks

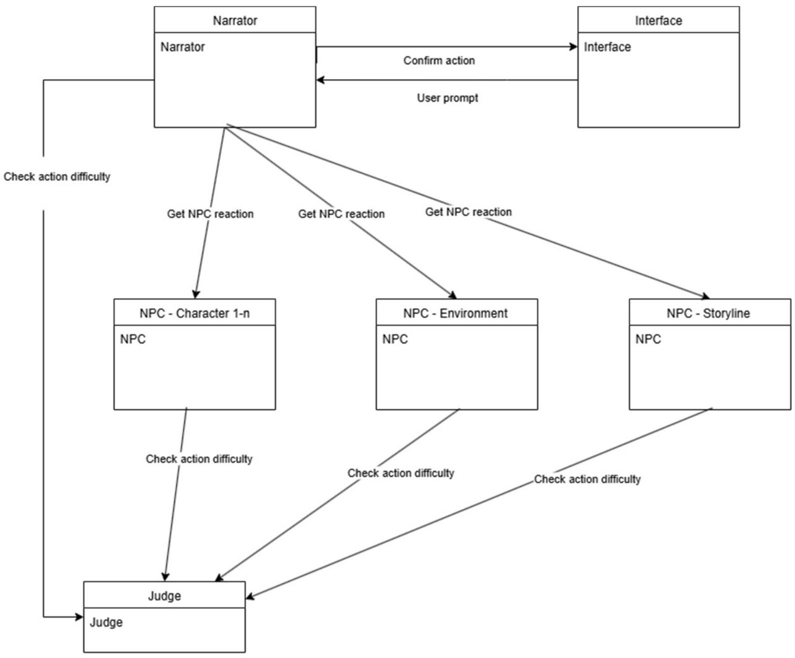


Game State
Retrieve game state infomation
Modify/add game state information
Difficulty
Get User decision
Display the action's difficulty
Difficulty
Interface
1.2
1.2.1
1.2.2
1.2.2.1
1.2.2.2
Narrator
1.1
1.1.1
1.1.1.1
1.1.2
1.1.3
User decision
Get user prompt
Prompt
Validate prompt's information
Extract user action
Action
Evaluate the action's difficulty
Prompt
Display the narratives
NPC reaction narrative
NPC reaction narrative
user action narrative
Judge
1.3
1.3.1
Retrieve rule's infomation
Rule
user action narrative
Generate NPC reaction narrative
Check Action result
Action result
Generate user action narrative
Result
Check reaction result
Difficulty
NPC
1.4
1.4.1
1.4.2
1.4.3
Evaluate the Reaction's difficulty
User action narrative
Reaction
Evaluate NPCs reaction
Retrieve Predefined NPC profile
Predefined NPC profile

1.4. Creating Agent Classes

1.4.1. Agent Description & Agent Class Diagram

There are four agents in this system to simulate the game master functionality. Here is the detailed description of those four agents defined in previous diagrams.



Interface Agent

- The interface agent displays the narrative generated by other agents.
- The interface agent displays necessary information, including main character (Health, Skill, Items), NPC (Health).
- The interface agent displays the difficulty of action (A number from 1 to 20, where a higher value indicates a more difficult action), and confirm the user's decision.
- The interface agent does the difficulty check (randomizes a number—success if it's higher than the difficulty number).

Narrator Agent

- Narrator agent generates main character narrative based on pre-defined story and user prompt
- Narrator agent generates new details or storyline based on user prompt if the not contain in pre-defined story
- Narrator agent generates enemy reaction narrative based on pre-defined story and NPC agents

- Narrator agent keeps track and manages the game state information (previous prompt and conversation, status of NPC, main character status, map, new-defined story).

Judge Agent

- Judge agent evaluates difficulty level of the action/reaction based on rule and reject unfeasible action/reaction.

NPC Agent

- NPC agent generates potential reaction to the main character or other NPC agents action based on the NPC profile (characteristic, status, background).

NPC Agent - Environment

- NPC agent - Environment generates potential reaction (trap, collapse, alarm) to the main character or other NPC agents action based on the pre-defined story.

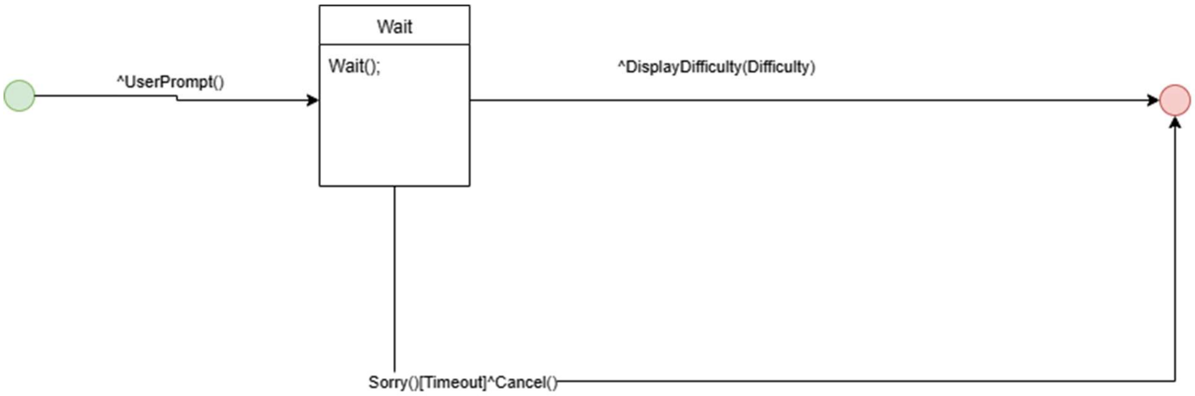
NPC Agent - Storyline

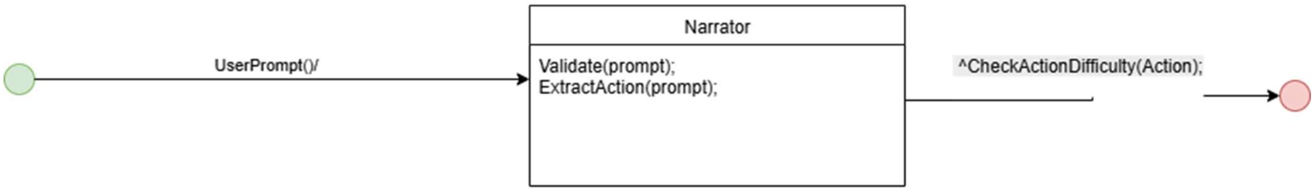
- NPC agent - Storyline generates potential impacts on the storyline based on the main character's actions and the pre-defined story.
- NPC agent - Storyline estimates the in-game time passed during the action of main character and other NPC.

1.5. Agent internal architecture

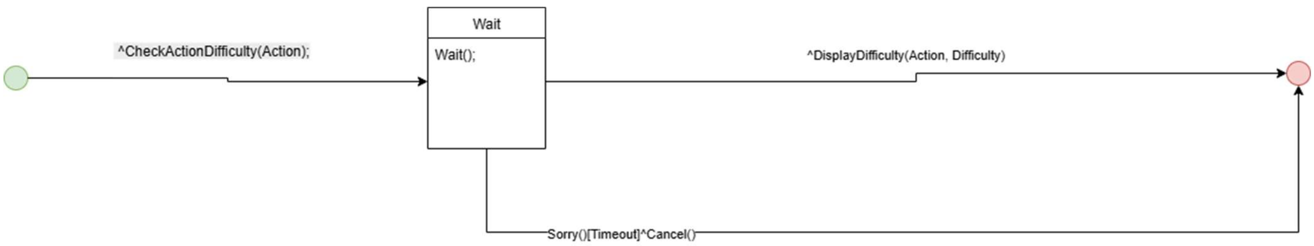
1.5.1. Conversations

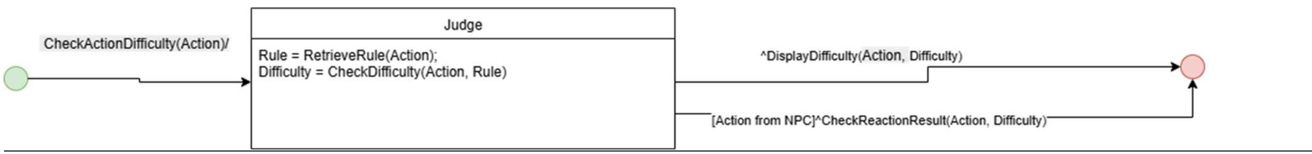
Get user prompt



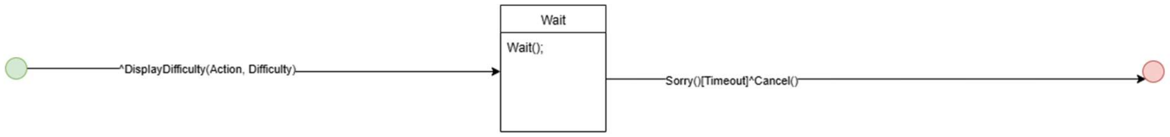


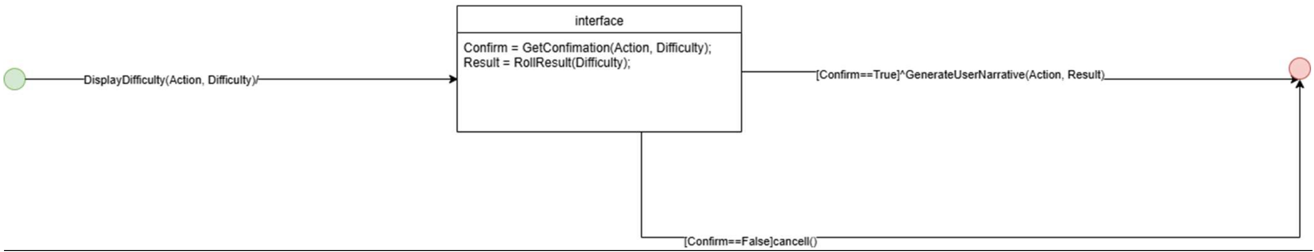
Check Action Difficulty



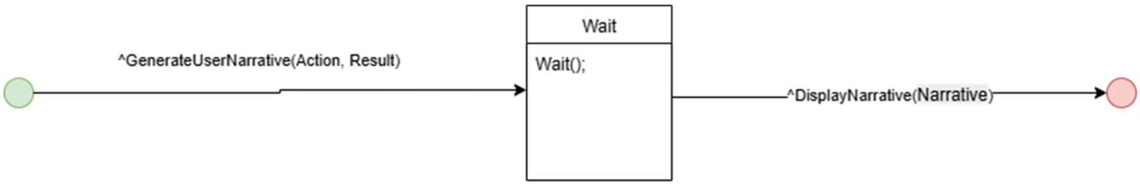


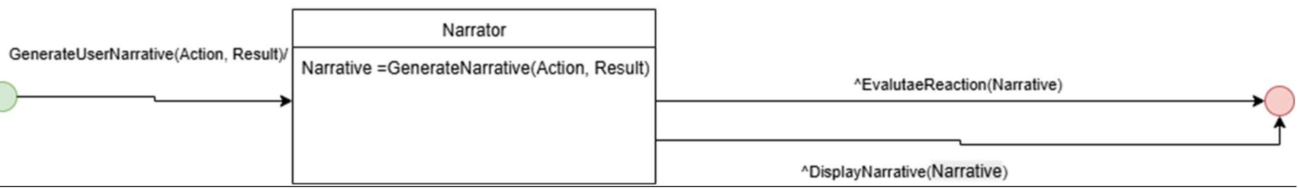
Display Difficulty



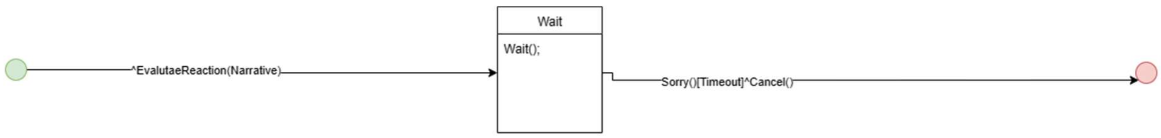


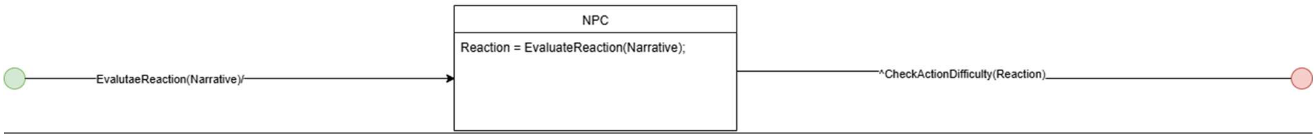
Generate User Narrative



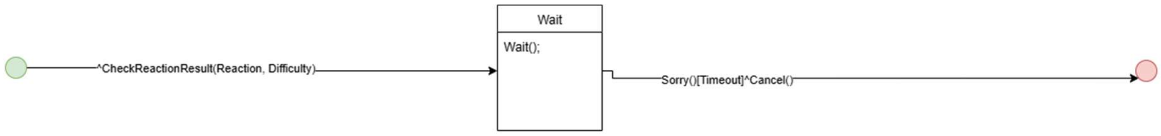


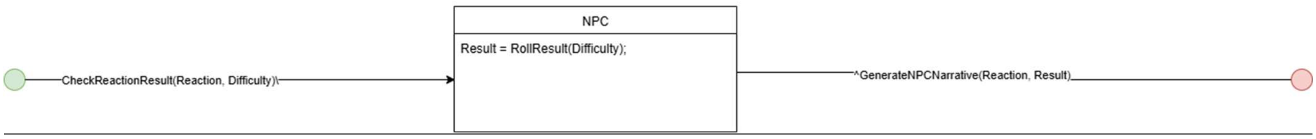
Evaluate Reaction



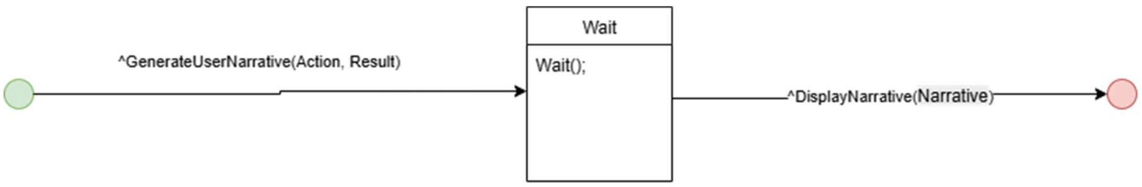


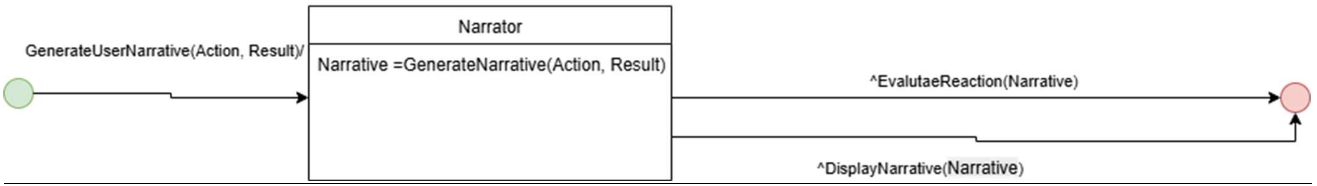
Check Reaction Result



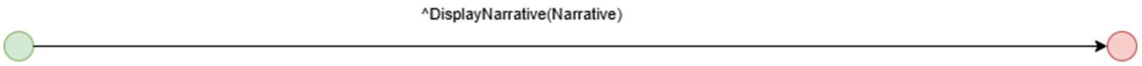


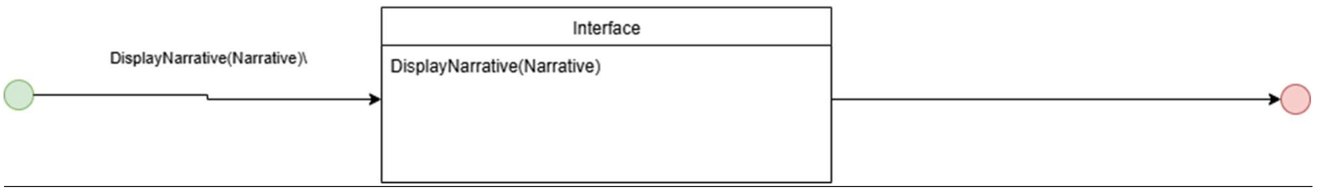
Generate NPC Narrative





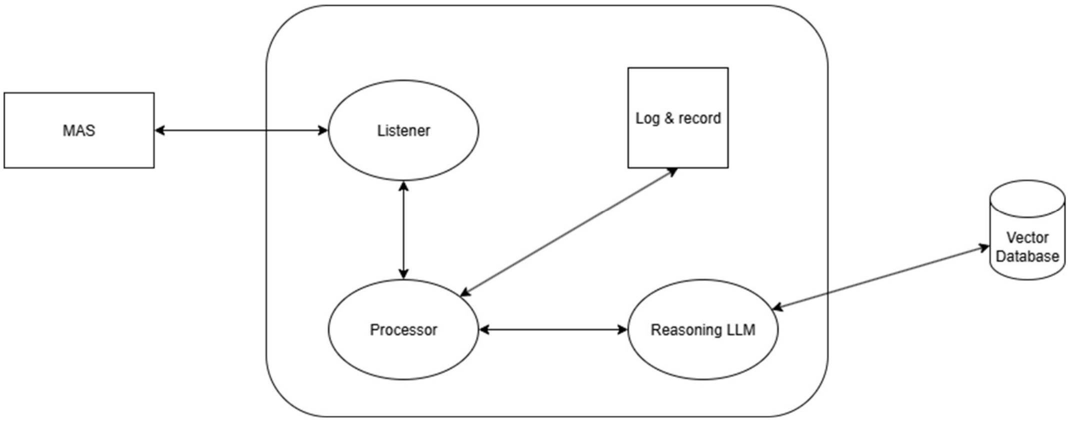
Display Narrative





1.5.2. Agent Architecture

Judge, NPC, Narrator:



Listener:

- Receive and sent all communication with other agents.

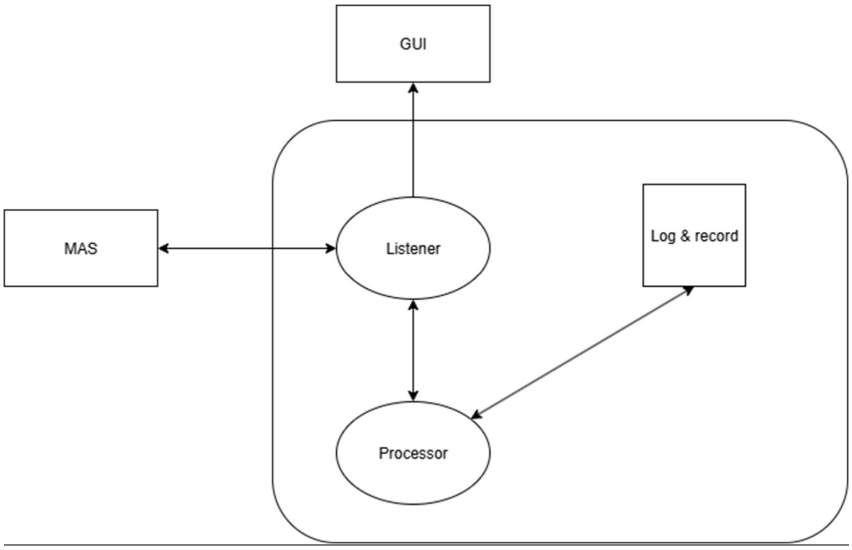
Processor:

- Pre-process and post-process the parameter to LLM
- Record and manage past record and the specific data. Input to LLM if necessary
    - NPC: Status of NPC
    - Narrator: Game State (previous prompt and conversation, main character status, map, status of NPC, new-defined story).

Reasoning LLM:

- Retrieve information from database based on task, and reasoning on the corresponding answer.

Interface:



Listener:

- Receive and sent all communication with other agents.
- Manage the GUI and its communication

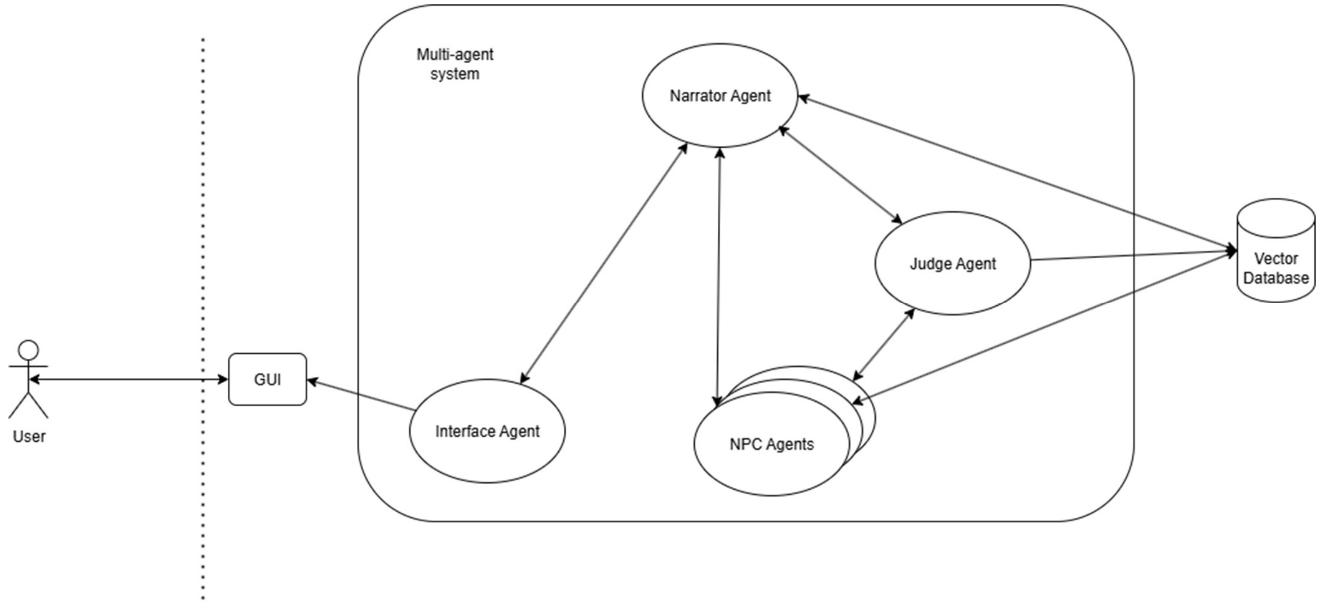
Processor:

- Record and manage past record.
- Process GUI logic if needed.

<u>Narrator:</u>

1.5.3. Deployment Diagrams

This system is intended to work cooperate with local GUI potentially in Java, and data (Rule, NPC, Items, pre-define story) in a vector database for RAG LLM potentially in crewAI.


User
GUI
Multi-agent system
Narrator Agent
Judge Agent
Interface Agent
NPC Agents
Vector Database